# TOC-2026: Quiz 5 Rubric (Reduction Procedure)

**Full Marks**: 10, **Time**: 30 minutes

**(1)** Consider the language $A_{TM} = \{\langle M, w\rangle | M \text{ is a Turing-Machine and } M \text{ accepts } w\}$. We describe the following procedure.

On input $\langle M, w\rangle$:

(i) Construct Turing-Machines $M_1$ and $M_2$ as follows

$M_1$ on input $x$

- If $(x \neq 101$ and $x \neq 010)$, then REJECT;
- Run $M$ on $w$;
- If ($M$ accepts $w$) then ACCEPT;
- If ($M$ does not accept $w$) then REJECT;

$M_2$ on input $y$

- If $y \neq \varepsilon$, then REJECT;
- Run $M$ on $w$;
- If ($M$ accepts $w$) then ACCEPT;
- If ($M$ does not accept $w$) then REJECT;

(ii) Output $\langle M_1, M_2\rangle$;

Consider the language $BTM = \{\langle M_1, M_2\rangle | M_1 \text{ and } M_2 \text{ are Turing-Machines such that } L(M_1) = \{101, 010\} \cdot L(M_2)\}$

Formally prove that $\langle M, w\rangle \in A_{TM}$ if and only if $\langle M_1, M_2\rangle \in BTM$ **(8 Marks)**

**Solution:** [Partial Proof]

**Proof:** [ $\implies$ ] Given that $\langle M, w\rangle \in A_{TM}$, we need to prove $\langle M_1, M_2\rangle \in BTM$.

From $\langle M, w\rangle \in A_{TM}$, we get that on input $w$, $M$ reaches ACCEPT state. Using this we examine the behavior of $M_1$ and $M_2$.

For $M_1$, we get that on an input string $x$, if $x \neq 101$ and $x \neq 010$, we reject $x$. Thus any string not in $\{101, 010\}$ is rejected. If $x = 101$ or $x = 010$ we accept them, since $w \in L(M)$, $M_1$ accepts $x$. Therefore, we get that $L(M_1) = \{101, 010\}$.

For $M_2$, we get that on an input string $y$, if $y \neq \varepsilon$, we reject $y$. Otherwise, since $w \in L(M)$, $M_2$ accepts $y$. Therefore, we get that $L(M_2) = \{\varepsilon\}$.

Examining $\langle M_1, M_2 \rangle$, we need to show that $L(M_1) = \{101, 010\} \cdot L(M_2)$. Since our sets have relatively, small size, we can calculate them by hand as $\{101, 010\} \cdot L(M_2) = \{101, 010\} \cdot \{\varepsilon\} = \{101\varepsilon, 010\varepsilon\} = \{101, 010\} = L(M_1)$. Therefore, we have $L(M_1) = \{101, 010\} \cdot L(M_2)$ and thus showing that $\langle M_1, M_2 \rangle \in BTM$.

[ $\Longleftarrow$ ] This part is **untrue** as given the fact that $\langle M, w \rangle \notin A_{TM}$, via the construction we get that $L(M_1) = L(M_2) = \varphi$. However, we still have that $L(M_1) = \{101, 010\} \cdot L(M_2) = \varphi$. Therefore, the backward direction is false. □

**(2)** Consider the language $D$ consisting of pairs of Turning-Machine encodings $\langle M_1, M_2 \rangle$ such that $L(M_1) \cap L(M_2)$ contains at least 20 strings. Formulate $D$ as a language in set-theoretic notation **(2 Marks)**

**Solution:** The set theoretic formulation is

$$D = \{\langle M_1, M_2 \rangle | M_1 \text{ and } M_2 \text{ are Turing-Machines such that } |L(M_1) \cap L(M_2)| \geq 20\}$$